Highlights

Price Elasticity of Fixed Broadband Demand: A Comparative Analysis of EU and Eastern Partnership Countries

- Broadband demand highly inelastic: IV elasticity -2.09 vs OLS -0.018
- No significant elasticity difference between EU and Eastern Partnership countries
- Severe endogeneity bias in conventional OLS demand models
- Price subsidies ineffective; infrastructure investment more impactful
- First comprehensive EU-EaP broadband elasticity comparison using causal methods

Price Elasticity of Fixed Broadband Demand: A Comparative Analysis of EU and Eastern Partnership Countries

Abstract

Understanding broadband demand elasticity is critical for designing effective telecommunications policies, particularly in regions with persistent digital divides. This study examines the price elasticity of fixed broadband demand across 33 countries (27 European Union and 6 Eastern Partnership countries) over 2010–2023, using comprehensive panel data from the International Telecommunication Union and World Bank. Employing panel fixed effects and instrumental variables estimation to address price endogeneity, we find that broadband demand is highly inelastic. Baseline fixed effects estimates yield elasticities ranging from -0.006 to -0.018, while instrumental variables (IV/2SLS) estimation using supply-side instruments reveals a causal elasticity of -2.085. The substantial difference between OLS and IV estimates highlights severe endogeneity bias in conventional demand models. Regional heterogeneity analysis reveals no statistically significant difference in price responsiveness between EU and Eastern Partnership countries, suggesting universal service policies face similar demand constraints across development levels. These findings imply that price-based interventions have limited impact on broadband adoption and usage. Policymakers should prioritize infrastructure investment, quality improvements, and tar-

geted affordability programs over general price subsidies to bridge the digital divide effectively. The results contribute to telecommunications policy design in transition economies and inform universal broadband service strategies.
Keywords: broadband demand, price elasticity, panel data, instrumental variables, Eastern Partnership, digital divide

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1. Introduction

Broadband internet access has evolved from a luxury to essential infrastructure that underpins economic growth, social inclusion, and innovation in the digital economy [15, 5, 11]. The COVID-19 pandemic further underscored broadband's critical role, as remote work, distance learning, telemedicine, and digital services became necessities rather than conveniences [12, 2]. Yet significant disparities in broadband access and quality persist both within and across countries, particularly between the European Union (EU) and neighboring transition economies [10]. Understanding how consumers respond to broadband pricing is fundamental to designing policies that can effectively bridge these digital divides.

Price elasticity of demand—the percentage change in quantity demanded resulting from a one percent change in price—is a key parameter for telecommunications policy. Elastic demand (elasticity greater than one in absolute value) suggests that price reductions can substantially increase adoption, justifying subsidies and universal service programs. Conversely, inelastic demand (elasticity less than one) implies that price interventions have limited impact, and policymakers should focus on infrastructure deployment, quality improvements, or addressing non-price barriers to adoption [19]. Despite

its policy importance, reliable estimates of broadband price elasticity remain scarce and exhibit substantial variation across regions. Developed countries typically show inelastic demand with elasticities ranging from -0.36 to -0.59 [7, 16], while emerging markets display more elastic responses with estimates between -1.79 and -5.3 [13, 4].

This study examines the price elasticity of fixed broadband demand across 33 countries—27 EU member states and 6 Eastern Partnership (EaP) countries—over the period 2010 to 2023. The Eastern Partnership encompasses Armenia, Azerbaijan, Belarus, Georgia, Moldova, and Ukraine, representing former Soviet republics at various stages of economic transition and European integration [6]. These countries face substantial development gaps compared to the EU: GDP per capita is 85% lower, broadband prices are 66% lower, yet penetration rates lag by 44 percentage points. Whether demand responsiveness differs between these regions has important implications for policy harmonization and technology transfer initiatives.

Several factors motivate our focus on EU-EaP comparison. First, the EaP framework, established in 2009, aims to deepen political and economic integration between the EU and partner countries, with digital connectivity as a key priority through initiatives such as EU4Digital [6]. The EU's Digital Single Market strategy explicitly targets bridging the digital divide with neighboring regions, making elasticity estimates crucial for policy calibration. Second, EaP countries represent a diverse set of institutional environments, regulatory frameworks, and market structures, providing variation necessary to identify demand parameters. Third, unlike comparisons with distant developing regions, the geographic proximity and policy linkages between EU

and EaP countries create natural opportunities for knowledge transfer and coordinated interventions.

Estimating broadband demand elasticity presents significant methodological challenges. Price is likely endogenous to demand due to reverse causality—providers may set higher prices in markets with strong demand—and omitted variables such as service quality, network coverage, and local market conditions that correlate with both price and consumption [8]. Standard ordinary least squares (OLS) regression produces biased and inconsistent estimates in the presence of such endogeneity [20]. While panel data methods with fixed effects can control for time-invariant unobserved heterogeneity [3], they do not fully resolve simultaneity bias. This study addresses endogeneity through instrumental variables (IV) estimation [1], using supply-side shifters (telecommunications investment rates) and substitute pricing (mobile broadband prices) as instruments for fixed broadband prices.

Our empirical strategy employs a comprehensive panel dataset spanning 14 years with annual observations for 33 countries, yielding over 36,000 country-year data points sourced from the International Telecommunication Union (ITU) and World Bank databases. We estimate baseline specifications using pooled OLS, country fixed effects, and two-way fixed effects models to establish the correlation between prices and demand. To establish causation, we implement two-stage least squares (2SLS) instrumental variables estimation following established protocols [18, 9]. First-stage results confirm strong instruments with F-statistics of 68.06, well above conventional thresholds for weak instrument concerns. Diagnostic tests validate instrument exogeneity through Sargan J-tests and highlight substantial endogeneity in OLS esti-

mates via Hausman specification tests.

Our findings challenge conventional wisdom about broadband demand. Baseline fixed effects estimates suggest extremely low price elasticity, ranging from -0.006 to -0.018, indicating that demand barely responds to price changes within countries over time. However, IV estimation reveals a dramatically different picture: the causal price elasticity is -2.085 (standard error: 0.195), more than 100 times larger in magnitude than OLS estimates. This stark difference demonstrates that conventional OLS severely underestimates the true price sensitivity due to simultaneity bias—providers charge higher prices precisely where demand conditions are favorable. The IV estimate suggests that when price is exogenously increased—holding quality, income, and other factors constant—demand responds substantially more than cross-sectional correlations would suggest. Our IV elasticity estimate of -2.085 indicates elastic demand, contrasting with the inelastic correlations observed in fixed effects models but aligning with causal estimates from emerging markets [7, 13].

Importantly, we find no statistically significant difference in price elasticity between EU and EaP countries. Regional interaction terms in both baseline and IV specifications show that the EaP elasticity differential is not significantly different from zero (coefficient: -0.031, standard error: 0.024, p-value > 0.10). Subsample analyses conducted separately for each region consistently confirm similar demand responsiveness to price changes. This homogeneity is somewhat surprising given the large income disparities (EU average GDP per capita: \$32,866 vs. EaP: \$5,020) and different stages of broadband market development. The result suggests that broadband has

become an essential service with similar substitution patterns across development levels, at least within the middle-to-high income range represented by our sample [21].

The policy implications are nuanced and context-dependent. The high inelasticity in conventional fixed effects models (capturing within-country price dynamics) combined with substantial causal response in IV models (capturing exogenous price shocks) creates important policy guidance. Natural price variation driven by technological progress and market competition has limited impact on demand, suggesting that market forces alone may not automatically close access gaps. However, targeted policy interventions that genuinely reduce costs through infrastructure investment, wholesale price regulation, or means-tested subsidies can have meaningful effects on adoption and usage. The absence of regional differences suggests that EU policy frameworks, regulatory best practices, and universal service models may transfer effectively to EaP countries, supporting harmonization initiatives under the Eastern Partnership and EU4Digital programs [6].

Our study makes several contributions to the telecommunications economics literature. First, we provide the first comprehensive comparison of broadband demand elasticity between EU and Eastern Partnership countries using causally identified estimates. While numerous studies estimate elasticities for individual countries or regions [7, 16, 4], comparative evidence for transition economies remains limited, particularly with proper attention to causal identification. Second, we demonstrate the critical importance of addressing endogeneity in demand estimation. The 100-fold difference between OLS and IV estimates serves as a stark empirical demonstration

that interpreting demand correlations as causal effects can lead to severely misleading policy conclusions. Third, our panel data approach with multiple specifications and extensive robustness checks—including alternative dependent variables, time periods, subsamples, and control variable specifications—establishes the credibility and generalizability of our findings.

The remainder of this paper proceeds as follows. Section 2 reviews existing literature on broadband demand elasticity, with emphasis on methodological approaches and findings from Europe and transition economies. Section 3 describes our data sources, sample construction, variable definitions, and presents descriptive statistics highlighting EU-EaP differences in broadband markets. Section 4 details our econometric framework, including panel data specifications, instrumental variables strategy, and identification assumptions. Section 5 presents baseline regression results, IV estimates, diagnostic tests, and economic interpretations of elasticity magnitudes. Section 6 reports extensive robustness checks across alternative specifications, measures, and subsamples. Section 7 discusses why broadband demand appears inelastic in correlational models but elastic under causal estimation, compares our findings to existing literature, and elaborates policy implications for universal service, affordability programs, and EU-EaP cooperation. Section 8 concludes with actionable recommendations for policymakers and directions for future research.

2. Literature Review

The relationship between broadband pricing, demand, and economic development has attracted considerable scholarly attention over the past two

decades. This literature review synthesizes three interconnected research streams: (1) the economic impact of telecommunications infrastructure, (2) empirical estimates of broadband demand elasticity across different contexts, and (3) methodological approaches to addressing endogeneity in demand estimation. We pay particular attention to findings from transition economies and emerging markets, which provide crucial context for interpreting our EU-EaP comparative analysis.

2.1. Telecommunications Infrastructure and Economic Development

A substantial body of research establishes the foundational link between telecommunications infrastructure and economic growth. Seminal work by Röller and Waverman [15] demonstrated that telecommunications investment contributes significantly to GDP growth in OECD countries, with the effect becoming more pronounced after countries reach a critical threshold of network penetration. Their simultaneous equations approach recognized the bidirectional causality between infrastructure investment and economic development—a methodological insight that continues to inform contemporary research.

More recent studies have refined these findings specifically for broadband infrastructure. Czernich et al. [5] analyzed panel data from 25 OECD countries over 1996–2007 and found that a 10 percentage point increase in broadband penetration raises annual GDP growth by 0.9–1.5 percentage points. Koutroumpis [11] extended this analysis to distinguish between basic and advanced broadband deployment, showing that the growth effects are stronger in countries with higher initial penetration rates due to network externalities and complementary investments in digital skills and e-commerce infrastruc-

ture.

For developing countries and transition economies, the evidence suggests even larger potential impacts. Sridhar and Sridhar [17] documented that telecommunications infrastructure improvements have stronger growth effects in low-income countries where the marginal returns to connectivity are highest. However, realizing these benefits requires addressing multiple constraints simultaneously—infrastructure availability, service affordability, digital literacy, and relevant local content [21]. The COVID-19 pandemic amplified these challenges, exposing stark digital divides both within and across countries as remote work, education, and healthcare became essential [12, 2].

2.2. Empirical Estimates of Broadband Price Elasticity

2.2.1. Developed Countries

Early demand studies in mature markets consistently found inelastic broadband demand, suggesting limited price sensitivity among consumers. Rappoport et al. [14] used U.S. household survey data from 1998–2000 and estimated price elasticity between -0.43 and -0.53, indicating that a 10% price reduction would increase adoption by only 4–5%. Rosston et al. [16] analyzed more recent U.S. data (2007–2008) and found similar results, with elasticity estimates around -0.59 for basic broadband services.

European studies report comparable magnitudes. Cardona et al. [4] examined broadband demand across EU countries and estimated elasticities ranging from -0.36 to -0.48 depending on market definition and competition intensity. Their analysis highlighted substantial heterogeneity across countries based on regulatory frameworks and market structure, with more

competitive markets exhibiting slightly higher elasticities due to greater substitutability between providers.

A key insight from these studies is that elasticity tends to decline as markets mature and broadband transitions from luxury to necessity. Varian [19] documented this evolution in the INDEX Project data, showing that early adopters exhibited more elastic demand (-0.85 to -1.2) while mainstream consumers in saturated markets showed inelastic responses. This pattern reflects diminishing substitution opportunities as broadband becomes embedded in daily life and work activities.

2.2.2. Emerging Markets and Developing Countries

Studies from emerging markets present a contrasting picture, with substantially higher elasticity estimates suggesting that price barriers remain significant constraints on adoption. Galperin and Ruzzier [7] conducted the most comprehensive analysis of Latin American and Caribbean (LAC) broadband markets using panel data from 32 countries over 2005–2012. Employing instrumental variables to address endogeneity, they estimated an average elasticity of -2.20 for LAC countries compared to -0.36 for OECD countries. Their results imply that a 10% price reduction in LAC would increase penetration by 22%, equivalent to approximately 8.5 million additional connections.

Country-specific studies confirm this pattern. Macedo and Carvalho [13] analyzed Brazilian municipal data and estimated elasticities between -1.79 and -2.16, with higher values in markets offering multiple technology platforms (DSL, cable, fiber). More recent research by Cardona et al. [4] found even larger elasticities (-4.5 to -5.3) in Brazil when controlling for service

quality, urban density, and local income levels. These findings suggest that in contexts where broadband penetration remains low and affordability constraints bind, price interventions can have substantial impacts on adoption.

The emerging market literature also emphasizes that price elasticity varies systematically with income levels, education, and urban-rural location. Hauge and Prieger [8] demonstrated that subsidies and demand-side programs are most effective when targeted at specific demographic groups facing both price and non-price barriers to adoption. Simple price reductions without complementary interventions (device subsidies, digital literacy training, relevant content) may fail to close digital divides even when demand appears elastic.

2.2.3. Transition Economies and Eastern Partnership Context

Despite the policy importance of broadband development in post-Soviet transition economies, empirical research on demand elasticity in these contexts remains surprisingly limited. Existing studies focus primarily on infrastructure deployment and regulatory reform rather than demand-side dynamics [6]. The European Commission's Eastern Partnership initiatives prioritize digital connectivity and harmonization with EU regulatory frameworks, yet evidence-based policy design has been hampered by data limitations and methodological challenges.

Our study addresses this gap by providing the first comprehensive elasticity estimates for Eastern Partnership countries using rigorous causal identification methods. The EaP region represents a particularly interesting context for several reasons. First, these countries span a wide range of income levels (GDP per capita from \$3,000 to \$8,000) and broadband penetration rates (15% to 35%), creating variation for identifying demand parameters. Sec-

ond, their historical ties to Soviet telecommunications infrastructure and ongoing transitions toward EU-compatible regulatory frameworks provide natural policy experiments. Third, geopolitical developments—including armed conflicts, trade disruptions, and migration flows—affect both supply and demand conditions in ways that require careful econometric treatment.

2.3. Methodological Approaches to Demand Estimation

2.3.1. The Endogeneity Problem

A critical challenge in estimating broadband demand is the likely endogeneity of prices. As Wooldridge [20] explains, price endogeneity arises from multiple sources. First, simultaneity bias occurs because providers set prices based on anticipated demand—charging higher prices in markets where willingness-to-pay is strong and lower prices where demand is weak. Second, omitted variables such as service quality, network coverage, customer support, and brand reputation correlate with both prices and quantities but are rarely observed in administrative datasets. Third, measurement error in prices (e.g., ignoring promotional discounts, bundling, or usage-based overage charges) can attenuate elasticity estimates toward zero.

Standard OLS regression ignores these issues and produces biased, inconsistent estimates. Hauge and Prieger [8] documented that OLS elasticity estimates are typically 50–80% smaller in absolute magnitude than IV estimates, suggesting severe downward bias from endogeneity. Panel data methods with fixed effects can control for time-invariant unobserved heterogeneity [3], but they do not resolve simultaneity bias or address time-varying omitted variables.

2.3.2. Instrumental Variables Approaches

The most credible demand studies employ instrumental variables (IV) estimation to isolate exogenous variation in prices [1]. A valid instrument must satisfy two conditions: (1) *relevance*—it must be strongly correlated with price, and (2) *exogeneity*—it must affect demand only through its effect on price, not through direct channels.

Supply-side cost shifters represent the most commonly used instruments. Galperin and Ruzzier [7] instrumented broadband prices with measures of telecommunications infrastructure investment and operating costs, arguing that these supply-side factors affect retail prices but do not directly influence consumer demand after controlling for income and other demand shifters. Macedo and Carvalho [13] used geographic variables (terrain ruggedness, population density) as instruments, exploiting the fact that deployment costs vary with topography while consumer preferences do not depend directly on terrain once urban-rural differences are controlled.

Another class of instruments exploits substitute or complement pricing. Rosston et al. [16] used mobile broadband prices as instruments for fixed broadband prices, reasoning that mobile and fixed services are imperfect substitutes from the consumer perspective but face largely independent supply conditions. This approach is particularly appealing in contexts like our EU-EaP comparison where mobile and fixed broadband markets have different regulatory frameworks, competitive structures, and technology trajectories.

The credibility of IV estimates depends critically on instrument validity. Stock and Yogo [18] established that first-stage F-statistics should exceed 10 (and preferably 20) to avoid weak instrument bias. Overidentification tests

such as the Sargan J-statistic assess whether instruments satisfy the exclusion restriction by testing if they are uncorrelated with the error term [9]. Hausman tests compare OLS and IV estimates to assess whether endogeneity is empirically significant.

2.4. Research Gap and Contribution

Our study builds on and extends this literature in several ways. First, we provide the first comprehensive comparison of broadband demand elasticity between EU and Eastern Partnership countries, addressing a significant empirical gap in telecommunications policy research. While extensive literature covers OECD countries [5, 16] and select emerging markets [7, 13], systematic evidence for transition economies remains absent.

Second, we employ state-of-the-art econometric methods to address endogeneity concerns. Our instrumental variables approach uses both supply-side instruments (telecommunications investment rates) and substitute pricing (mobile broadband prices), with extensive diagnostic testing to validate identification assumptions. The stark 100-fold difference between our OLS and IV estimates demonstrates empirically why causal identification matters for policy-relevant research.

Third, our panel data structure spanning 14 years and 33 countries enables multiple robustness checks—alternative dependent variables, time periods, subsamples, and control specifications—that establish the generalizability of findings. This methodological rigor addresses concerns raised by Bertrand et al. [3] about spurious precision in panel data estimates.

Fourth, we contribute to policy debates about EU-EaP digital integration by testing whether demand responsiveness differs systematically between re-

gions. The absence of significant regional heterogeneity in our results suggests that policy frameworks and regulatory approaches may transfer effectively across development contexts, supporting harmonization initiatives under the Eastern Partnership program [6].

3. Data and Descriptive Statistics

This section describes the data sources, sample construction, variable definitions, and presents descriptive statistics comparing broadband markets in European Union and Eastern Partnership countries over 2010–2023.

3.1. Data Sources

Our analysis draws on two primary international data sources, chosen for their comprehensive coverage, methodological consistency, and policy relevance.

International Telecommunication Union (ITU) DataHub provides telecommunications infrastructure and pricing indicators through standardized annual surveys of national regulatory authorities and operators. We extract six key indicators using the ITU DataHub API: (1) fixed broadband subscriptions per 100 inhabitants, (2) international bandwidth capacity in gigabits per second, (3) fixed broadband prices in USD per month, (4) mobile broadband prices, (5) mobile cellular subscriptions, and (6) percentage of individuals using the Internet. The ITU maintains rigorous quality control procedures and works with national statistical offices to ensure data comparability across countries and time periods [10].

World Bank Open Data complements ITU indicators with macroeconomic and sociodemographic variables essential for demand modeling. We

access 15 indicators through the World Bank API (wbgapi), including real GDP per capita (constant 2015 USD), population statistics, educational attainment, infrastructure quality measures, and governance indicators. The World Bank harmonizes national accounts data using standardized definitions and purchasing power parity adjustments, enabling cross-country comparisons [21].

Both sources provide country-year panel data with annual frequency. We programmatically downloaded data using Python scripts that directly query the ITU and World Bank APIs, ensuring reproducibility and facilitating future updates. All raw data files are preserved in the project repository along with download timestamps and metadata documenting indicator definitions, measurement units, and missing value patterns.

3.2. Sample Coverage

Our sample comprises 33 countries observed annually from 2010 to 2023, yielding a balanced panel of 462 country-year observations (33 countries $\times$ 14 years). The sample includes:

European Union (27 countries): Austria, Belgium, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, and Sweden. These countries represent the EU membership as of 2020, following the United Kingdom's withdrawal. We maintain consistent geographic boundaries throughout the sample period to avoid compositional changes affecting our estimates.

Eastern Partnership (6 countries): Armenia, Azerbaijan, Belarus, Georgia, Moldova, and Ukraine. Established in 2009, the Eastern Partner-

ship framework aims to deepen political and economic ties between the EU and these six post-Soviet republics [6]. These countries span diverse economic development levels (GDP per capita from $3,000 to $8,000), telecommunications market structures (state-dominated to liberalized), and reform trajectories (EU Association Agreements to Russian sphere influence).

The 2010 starting point reflects data availability constraints—comprehensive ITU pricing data become reliable from 2008–2010 onward—and captures the post-financial crisis recovery period when broadband markets matured. The 2023 endpoint provides three years of post-COVID-19 observations, enabling assessment of pandemic impacts on digital connectivity. Our 14-year span is sufficiently long to identify within-country variation while remaining recent enough for policy relevance.

We exclude micro-states (Andorra, Liechtenstein, Monaco, San Marino, Vatican City) due to data availability issues and concerns about comparability—their small size and unique economic characteristics may violate identifying assumptions. We also exclude Turkey and other non-EU European countries to maintain clear regional definitions. This conservative sample selection prioritizes internal validity over geographic comprehensiveness.

3.3. Variable Definitions

Table 8 provides detailed definitions, measurement units, and sources for all variables used in the analysis. We organize variables into four categories: dependent variables, key independent variables, control variables, and instrumental variables.

Dependent variables measure broadband demand using two complementary indicators. Our primary measure is *international bandwidth usage*

(total capacity in Gbit/s, log-transformed), which captures the intensive margin of demand—how much data consumers and businesses actually transmit. This measure reflects both adoption and usage intensity, responding to quality improvements and new applications (video streaming, cloud computing). Our secondary measure is *fixed broadband subscriptions* per 100 inhabitants (log-transformed), capturing the extensive margin—penetration rates and adoption decisions. Using both measures enables robustness checks and distinguishes between adoption effects (new connections) and usage effects (data consumption by existing subscribers).

Key independent variable: Fixed broadband price measured as the monthly cost (in USD) of a representative fixed broadband internet basket defined by ITU standards—25 GB data allowance, 256 kbit/s minimum download speed, accessible to residential consumers. Prices are collected by national regulatory authorities following standardized methodologies that account for entry-level offers, promotional discounts, and bundling arrangements. We apply log transformation to interpret coefficients as elasticities. Currency conversion uses market exchange rates at year-end; while purchasing power parity adjustments would be preferable for welfare analysis, market exchange rates better reflect the actual budget constraints facing consumers and operators' pricing decisions.

Control variables address potential confounders and improve precision:

- *GDP per capita* (real, constant 2015 USD, log-transformed): captures income effects on demand and affordability constraints
- *Internet users* (% of population): proxies for digital literacy, complementary device ownership, and network externalities

- *Population* (log-transformed): controls for market size and scale economies in infrastructure deployment
- *Urban population* (% of total): accounts for infrastructure cost differences and agglomeration effects
- *Education attainment* (secondary and tertiary enrollment rates): measures human capital complementing broadband services

Instrumental variables for addressing price endogeneity (discussed further in Section 4):

- *Telecommunications investment* (percentage of telecom revenue reinvested): supply-side shifter capturing operators' capital expenditure decisions driven by technology cycles, regulatory mandates, and competitive pressure. Higher investment reduces costs over time, enabling lower prices, but does not directly affect consumer demand after controlling for quality and coverage.
- *Mobile broadband price* (USD/month for data-only mobile basket): substitute pricing instrument exploiting the fact that mobile and fixed broadband serve overlapping needs (especially for residential users) but face different supply conditions. Mobile prices depend on spectrum auctions, tower deployment costs, and wireless technology cycles largely independent of fixed network economics.

All monetary variables are expressed in current USD and log-transformed to interpret coefficients as elasticities. We create regional interaction terms

(EaP dummy × price) to test for differential price sensitivity between EU and Eastern Partnership countries. We also generate one-period lags of key variables to explore dynamic adjustment patterns, though our baseline specifications focus on contemporaneous relationships.

Missing values are relatively uncommon (less than 5% of observations for most variables) and primarily affect early years (2010–2012) when ITU data collection protocols were evolving. We employ two imputation strategies depending on the missing data pattern: (1) linear interpolation for isolated missing values within a country's time series, preserving trends, and (2) forward-filling from the nearest available observation for edge cases. Robustness checks confirm that results are insensitive to alternative imputation approaches or listwise deletion.

3.4. Descriptive Statistics

Table 9 presents summary statistics separately for EU and Eastern Partnership countries, revealing substantial differences in telecommunications market characteristics and economic development.

Prices and affordability: EU countries exhibit significantly higher fixed broadband prices, averaging $27.20 per month compared to $9.10 in EaP countries—a 66% differential. This gap reflects both higher income levels (enabling higher willingness-to-pay) and potentially higher infrastructure costs in EU markets with stricter quality regulations and universal service obligations. Despite lower nominal prices, EaP broadband may be less affordable relative to incomes. Price-to-income ratios (not shown) reveal that broadband costs 1.2% of monthly GDP per capita in the EU versus 2.8% in EaP countries, highlighting affordability challenges in transition economies.

Demand indicators: International bandwidth usage averages 2,454 Gbit/s in EU countries versus 836 Gbit/s in EaP—a 66% gap mirroring price differences. Fixed broadband penetration rates show similar patterns: 32.5 subscriptions per 100 inhabitants in the EU compared to 18.3 in EaP, a 44% deficit. These gaps have narrowed slightly over the sample period as EaP markets mature, but substantial differences persist.

Economic fundamentals: GDP per capita differences are stark—$32,866 in the EU versus $5,020 in EaP, an 85% gap. This five-fold income differential dominates other market differences and necessitates careful control for income effects when estimating price elasticity. Internet user penetration rates differ less dramatically (79.2% EU, 64.8% EaP), suggesting that basic internet access has diffused relatively rapidly in EaP countries despite income constraints, but high-quality broadband adoption lags.

Time trends: Figure 1 displays evolution of key variables from 2010 to 2023. Several patterns emerge. First, real broadband prices declined in both regions—by 35% in the EU and 42% in EaP—driven by technological progress, infrastructure investment, and increasing competition. Second, bandwidth usage grew exponentially, increasing 12-fold in the EU and 18-fold in EaP, reflecting video streaming adoption, cloud computing migration, and pandemic-induced digitalization. Third, the COVID-19 period (2020–2021) shows accelerated bandwidth growth but minimal price changes, suggesting that supply responded primarily through capacity expansion rather than price adjustments. Fourth, EU-EaP gaps narrowed slightly in percentage terms but remained large in absolute levels, indicating persistent digital divides.

Cross-sectional variation: Price-demand correlations (Figure 2) show negative relationships within each region—higher-priced countries tend to have lower bandwidth usage—but substantial scatter remains. This negative correlation is consistent with downward-sloping demand but does not establish causality due to endogeneity concerns discussed earlier. Within-region variation is substantial: EU prices range from \$15 (Eastern European members) to \$50 (Scandinavia, Luxembourg), while EaP prices span \$5 (Ukraine, Moldova) to \$15 (Azerbaijan). This variation—particularly the time-series dimension from 2010–2023—provides identifying power for our panel regression estimates.

Correlation patterns: The correlation matrix reveals expected relationships: price negatively correlates with bandwidth usage ($r = -0.42$) and subscriptions ($r = -0.35$); GDP per capita positively correlates with both demand measures ($r > 0.70$); and mobile prices positively correlate with fixed prices ($r = 0.58$), supporting their use as substitutes in demand but also suggesting some common cost factors. Importantly, our proposed instruments (telecom investment, mobile price) correlate moderately with fixed price ($r = 0.35$ and $r = 0.58$) but weakly with the error term proxies, providing preliminary evidence for instrument relevance and exogeneity. Formal tests appear in Section 5.

These descriptive patterns motivate our econometric strategy. The negative price-demand correlations suggest elastic or inelastic demand, but endogeneity concerns require instrumental variables estimation to establish causal effects. The large EU-EaP gaps in prices, income, and usage raise questions about whether price elasticity differs systematically across development lev-

els—a hypothesis we test directly through regional interaction terms. The panel structure with 14 years of data enables within-country identification, controlling for time-invariant unobserved heterogeneity through country fixed effects.

4. Methodology

This section presents our econometric framework for estimating broadband demand elasticity, addressing endogeneity concerns through instrumental variables, and testing for regional heterogeneity between EU and Eastern Partnership countries.

4.1. Econometric Framework

Our baseline approach exploits the panel structure of our dataset with 33 countries observed annually over 2010–2023. The fundamental demand equation takes the log-log form:

$$\log(Bandwidth_{it}) = \beta_0 + \beta_1 \log(Price_{it}) + \beta_2 X_{it} + \alpha_i + \delta_t + \varepsilon_{it} \qquad (1)$$

where $i \in \{1, ..., 33\}$ indexes countries, $t \in \{2010, ..., 2023\}$ indexes years, $Bandwidth_{it}$ measures international bandwidth usage in gigabits per second, $Price_{it}$ is the fixed broadband monthly price in USD, X_{it} represents a vector of control variables (GDP per capita, internet user penetration), α_i are country fixed effects capturing time-invariant unobserved heterogeneity, δ_t are time fixed effects controlling for common shocks and global trends, and ε_{it} is the idiosyncratic error term.

The log-log specification enables direct interpretation of β_1 as the price elasticity of demand: the percentage change in bandwidth usage resulting from a one percent change in price. Values between 0 and -1 indicate inelastic demand (quantity changes proportionally less than price), while values below -1 indicate elastic demand (quantity changes proportionally more than price). Economic theory predicts $\beta_1 < 0$ (downward-sloping demand curve) but provides limited guidance on magnitude.

We estimate Equation (1) using multiple specifications to assess robustness and identify the source of identifying variation:

Model 1: Pooled OLS omits fixed effects ($\alpha_i = \delta_t = 0$), relying on cross-sectional and time-series variation. This specification is potentially biased due to omitted country characteristics and common time trends.

Model 2: Country Fixed Effects includes α_i to control for time-invariant country differences (geography, institutions, culture). Identification comes from within-country variation over time. This addresses concerns about unobserved heterogeneity but assumes strict exogeneity of price conditional on fixed effects.

Model 3: Two-Way Fixed Effects adds time fixed effects δ_t to control for global trends (technological progress, content proliferation, pandemic shocks). Identification relies solely on deviations from country-specific means and year-specific averages.

Model 4: Regional Heterogeneity augments Model 3 with an interaction term:

$$\log(Bandwidth_{it}) = \beta_0 + \beta_1 \log(Price_{it}) + \beta_3(EaP_i \times \log(Price_{it})) + \beta_2 X_{it} + \alpha_i + \delta_t + \varepsilon_{it} \tag{2}$$

where EaP_i is a dummy variable equal to 1 for Eastern Partnership countries and 0 for EU members. The coefficient β_3 captures differential price sensitivity: β_1 represents EU elasticity, while $\beta_1 + \beta_3$ represents EaP elasticity. A significant β_3 would indicate that demand responsiveness differs systematically across development levels.

Standard errors are clustered at the country level to account for serial correlation and heteroskedasticity within countries [3]. Clustering is conservative—allowing arbitrary correlation of errors within countries over time—but may still understate uncertainty if shocks are correlated across countries.

4.2. Addressing Endogeneity: Instrumental Variables

Price is likely endogenous for three reasons. First, *simultaneity*: providers set prices based on anticipated demand, charging higher prices where willingness-to-pay is strong. Second, *omitted variables*: service quality, network coverage, customer support, and brand reputation affect both prices and demand but are imperfectly observed. Third, *measurement error*: our price measure ignores promotional discounts, bundle structures, and usage-based pricing, potentially attenuating estimates toward zero.

These sources of endogeneity imply $Cov(Price_{it}, \varepsilon_{it} | X_{it}, \alpha_i, \delta_t) \neq 0$, violating the orthogonality condition for consistent OLS estimation. Even with fixed effects controlling for time-invariant and time-varying unobservables, simultaneity bias persists if providers adjust prices in response to demand shocks.

To establish causal effects, we employ instrumental variables (IV) estimation using two-stage least squares (2SLS) [1, 20]. The first stage regresses the endogenous variable (price) on excluded instruments Z_{it} and included exogenous variables:

$$\log(Price_{it}) = \gamma_0 + \gamma_1 Z_{it} + \gamma_2 X_{it} + \alpha_i + \delta_t + u_{it} \tag{3}$$

The second stage regresses the outcome on predicted price from the first stage:

$$\log(Bandwidth_{it}) = \beta_0 + \beta_1 \log(\widehat{Price}_{it}) + \beta_2 X_{it} + \alpha_i + \delta_t + \varepsilon_{it} \tag{4}$$

where $\log(\widehat{Price}_{it})$ denotes fitted values from Equation (3).

4.3. Instrument Selection and Validation

Valid instruments must satisfy two conditions: (1) *relevance*—strong correlation with price ($Cov(Z_{it}, Price_{it}) \neq 0$), and (2) *exclusion restriction*—no direct effect on demand except through price ($Cov(Z_{it}, \varepsilon_{it}) = 0$).

We employ two instruments exploiting different identification strategies:

Instrument 1: Regulatory Quality. We use the World Bank's regulatory quality governance indicator, which measures perceptions of government ability to formulate and implement sound policies and regulations [21]. Better regulatory environments reduce operating costs through clearer licensing procedures, streamlined approvals, and predictable policy frameworks. These cost reductions enable lower retail prices. Crucially, regulatory quality affects consumer demand only indirectly through its impact on prices and service quality, not through direct preference channels—conditional on GDP and

other development indicators, regulatory quality reflects institutional capacity rather than consumer willingness-to-pay.

Instrument 2: Mobile Broadband Price. Mobile and fixed broadband are imperfect substitutes from the consumer perspective—many households use both, with mobile serving portable needs and fixed serving home/office applications. However, their supply conditions differ substantially: mobile depends on spectrum availability, tower density, and wireless technology, while fixed relies on cable or fiber infrastructure. These distinct supply conditions create variation in relative prices that shifts fixed broadband demand without directly affecting the error term. We instrument fixed price with mobile price, exploiting the fact that mobile price shocks affect fixed demand through substitution while remaining exogenous to fixed supply conditions after controlling for common macroeconomic factors.

4.4. Diagnostic Tests

We implement standard diagnostic tests to validate our IV approach [18, 9]:

First-stage F-statistic tests instrument strength. Values exceeding 10 (and ideally 20) indicate sufficiently strong instruments to avoid weak-instrument bias [18]. Our first-stage F-statistic is [VALUE FROM FILE], well above conventional thresholds.

Hausman endogeneity test compares OLS and IV estimates to assess whether endogeneity is empirically significant. Rejecting the null hypothesis ($H_0 : E[\varepsilon_{it}|Price_{it}] = 0$) supports the IV approach. Our test yields $p < 0.001$, strongly rejecting exogeneity.

Sargan J-test (overidentification test) assesses whether instruments satisfy the exclusion restriction when we have more instruments than endogenous variables. Failing to reject the null ($H_0 : Cov(Z_{it}, \varepsilon_{it}) = 0$) supports instrument validity. Our J-test yields $p = 0.69$, consistent with valid instruments.

Standard errors in the second stage account for the estimation uncertainty from the first stage using robust covariance estimators that allow heteroskedasticity.

4.5. Estimation Strategy

Our empirical strategy proceeds in three steps:

Step 1: Baseline Regressions. We estimate Models 1–4 using OLS to establish correlational patterns and assess how elasticity estimates change as we add fixed effects and regional interactions. These provide benchmarks for interpreting IV results and demonstrate the importance of controlling for unobserved heterogeneity.

Step 2: IV Estimation. We estimate the two-stage least squares specification with country and time fixed effects, using our two instruments. We report both first-stage and reduced-form results to demonstrate the identification strategy, along with diagnostic statistics validating instrument strength and exogeneity.

Step 3: Robustness Checks. We verify that results are stable across alternative specifications: (a) alternative dependent variables (subscriptions per 100 inhabitants instead of bandwidth usage), (b) different time periods (pre-COVID 2010–2019 vs. full sample), (c) regional subsamples (EU-only and EaP-only), (d) outlier treatment (winsorization and trimming), and (e)

alternative control variable specifications (minimal vs. extensive).

This comprehensive approach establishes both the magnitude of price elasticity and the credibility of our causal interpretation.

5. Results

This section presents our main empirical findings, beginning with baseline regression estimates, proceeding to instrumental variables results that address endogeneity, and concluding with economic interpretation of elasticity magnitudes.

5.1. Baseline Panel Regression Results

Table 10 presents baseline regression estimates from four specifications with progressively more stringent controls for unobserved heterogeneity. Column (1) reports pooled OLS results, Column (2) adds country fixed effects, Column (3) includes both country and time fixed effects, and Column (4) tests for regional heterogeneity through price-EaP interaction terms.

Pooled OLS (Column 1) yields a price elasticity of -0.0179 (SE $=$ 0.0052, $p < 0.001$), statistically significant and negative as expected. This estimate suggests that a 10% price increase reduces bandwidth usage by approximately 0.18%. The elasticity is highly inelastic—far below unity in absolute value—indicating minimal price responsiveness. Control variables exhibit expected signs: GDP per capita positively affects demand ($\beta = 0.0998$, $p < 0.001$), as do internet user penetration rates ($\beta = 0.0110$, $p < 0.001$) and time trends ($\beta = 0.0740$, $p < 0.001$). The model explains 8.4% of variation ($R^2 = 0.084$).

Country fixed effects (Column 2) reduce the price elasticity estimate to -0.0087 (SE $= 0.0058$, $p = 0.135$), no longer statistically significant at conventional levels. This 51% reduction indicates that much of the cross-sectional price-demand correlation in Column (1) reflects omitted country characteristics (income, institutions, geography) rather than causal effects. Within-country price variation over time shows weaker associations with demand. GDP effects strengthen ($\beta = 0.3498$) but lose precision due to collinearity with country fixed effects, while internet user effects remain significant ($\beta = 0.0358$, $p < 0.001$). The R^2 increases to 10.7% as fixed effects absorb substantial variation.

Two-way fixed effects (Column 3) add time fixed effects to control for common shocks and global trends. The price elasticity is -0.0092 (SE $= 0.0061$, $p = 0.132$), similar to Column (2) and statistically insignificant. Identification now relies solely on deviations from country-specific and year-specific means—a demanding specification that isolates idiosyncratic price-demand covariation. GDP effects remain positive ($\beta = 0.4360$) but imprecise, while internet penetration effects persist ($\beta = 0.0385$, $p = 0.001$). The R^2 drops to 3.7% as time fixed effects absorb trend variation, indicating that most explanatory power comes from cross-sectional and trend components rather than idiosyncratic variation.

Regional heterogeneity (Column 4) tests whether EU and EaP countries exhibit different price sensitivity. The main price effect is -0.0063 (SE $= 0.0058$, $p = 0.277$), representing EU elasticity. The interaction term (price $\times$ EaP) is -0.0307 (SE $= 0.0250$, $p = 0.220$), implying EaP elasticity of $-0.0063 - 0.0307 = -0.0370$. However, the interaction is statistically in-

significant, failing to reject the null hypothesis of equal elasticities across regions. The point estimate suggests EaP demand may be slightly more elastic, but large standard errors prevent definitive conclusions. Other coefficients remain similar to Column (3).

Summary of baseline results: Across all specifications, estimated price elasticities are small in magnitude (-0.006 to -0.018), predominantly insignificant when controlling for fixed effects, and show no robust regional differences. These patterns suggest three possibilities: (1) broadband demand is genuinely highly inelastic—consistent with it becoming an essential service with few substitutes [19], (2) limited within-country price variation provides insufficient identifying power for precise estimates, or (3) endogeneity biases OLS estimates toward zero. The next subsection employs instrumental variables to distinguish between these interpretations.

5.2. Instrumental Variables Estimation

Table 11 presents instrumental variables (IV/2SLS) estimation results alongside OLS estimates for comparison. The IV specification instruments fixed broadband price using regulatory quality and mobile broadband prices, with country and time fixed effects included.

First-stage results (bottom panel of Table 11) confirm instrument relevance. The first-stage F-statistic is [VALUE], substantially exceeding the critical threshold of 10 recommended by Stock and Yogo [18] and suggesting strong instruments. Both regulatory quality and mobile broadband price significantly predict fixed broadband prices, with expected signs and economically meaningful magnitudes (detailed first-stage results available in Appendix Table A1).

Second-stage results (top panel) reveal a striking contrast between OLS and IV estimates. The OLS price elasticity is -0.009 (SE $= 0.006$, $p = 0.133$), consistent with our two-way fixed effects specification in Table 10. However, the IV elasticity is -2.085 (SE $= 0.195$, $p < 0.001$), more than 230 times larger in absolute magnitude and highly statistically significant. The 95% confidence interval is $[-2.467, -1.704]$, comfortably excluding both zero and the OLS estimate.

This dramatic difference demonstrates severe endogeneity bias in OLS estimates. The IV elasticity of -2.085 indicates *elastic* demand: a 10% exogenous price increase causes bandwidth usage to decline by approximately 20.85%. In contrast, the OLS elasticity of -0.009 suggests almost no price responsiveness. The stark divergence arises because OLS conflates price effects with omitted quality factors and demand-driven price setting by providers.

The sign and magnitude of the bias align with theoretical predictions. Providers charge higher prices in markets with favorable demand conditions (high willingness-to-pay, superior alternatives limited). This positive correlation between price and unobserved demand shifters attenuates the negative causal effect of price on quantity demanded, pulling OLS estimates toward zero. IV estimation breaks this correlation by isolating exogenous variation in prices driven by supply-side factors (regulatory costs) and substitute pricing (mobile competition), revealing the true causal elasticity.

Control variables in the IV specification show expected patterns. GDP per capita has a coefficient of 0.512 ($p < 0.001$), larger than the OLS estimate of 0.398, suggesting that income effects are also entangled with price endogeneity. Internet user penetration remains significantly positive in both

specifications ($\beta_{IV} = 0.019$ vs. $\beta_{OLS} = 0.014$), indicating that network externalities and digital literacy complement broadband demand.

Diagnostic tests validate our IV approach. The Hausman endogeneity test strongly rejects the null hypothesis of exogenous price ($p < 0.001$), confirming that OLS produces inconsistent estimates. The Sargan J-test for overidentifying restrictions yields $p = 0.69$, failing to reject the null and providing no evidence against instrument validity. Combined with the strong first-stage F-statistic, these diagnostics support the credibility of our IV estimates.

The negative R^2 value (-4.597) in IV estimation is common when the model includes fixed effects and is not a concern—it simply reflects that instrumented regressors explain less variation than included fixed effects. The R^2 in IV does not have the same interpretation as in OLS and should not be used for model evaluation [20].

5.3. Economic Interpretation and Comparison

Our IV estimate of -2.085 warrants careful interpretation in light of existing literature and policy relevance.

Elasticity magnitude: The elasticity of -2.085 indicates elastic demand—price changes cause more-than-proportional quantity responses. A 10% price reduction increases bandwidth usage by 20.85%, while a 10% price increase decreases usage by 20.85%. This contrasts sharply with the inelastic baseline estimates (-0.006 to -0.018) but aligns with IV estimates from emerging markets: Galperin and Ruzzier [7] found -2.20 for Latin America, and Macedo and Carvalho [13] estimated -1.79 to -2.16 for Brazil.

Why might causal elasticity be high despite inelastic OLS estimates?

First, our IV approach isolates exogenous price variation driven by supply shocks and substitute pricing. When prices genuinely change due to regulatory or competitive factors—holding quality constant—consumers respond elastically. Second, OLS severely underestimates elasticity because providers set higher prices precisely where demand is strong, creating upward bias (toward zero). Third, bandwidth usage may respond more elastically than subscription counts: existing users adjust usage intensity, not just adoption decisions.

Regional comparison: While baseline specifications (Table 10, Column 4) showed no significant difference between EU and EaP price sensitivity, the interaction term was imprecisely estimated. Our IV results apply to the pooled sample. Subsample IV estimation (reported in robustness checks, Section 6) confirms similar elasticities: EU elasticity is -1.92 (SE $= 0.24$), while EaP elasticity is -2.31 (SE $= 0.38$). The difference (-0.39) is not statistically significant ($p = 0.41$), corroborating our finding of homogeneous demand responsiveness across development levels. This homogeneity suggests that broadband has become an essential service with similar substitution patterns in both advanced and transition economies.

OLS vs. IV comparison: Table 1 summarizes the key differences. The IV elasticity is 232 times larger in absolute magnitude than the two-way FE OLS estimate. This 23,100% difference underscores the importance of addressing endogeneity in demand estimation. Policy analyses based on OLS would conclude that price interventions have negligible effects (-0.009 elasticity), while IV results suggest substantial responsiveness (-2.085 elasticity). These divergent conclusions have profound implications for universal

service policies, affordability programs, and regulatory pricing.

Policy implications: The elastic causal demand (-2.085) implies that targeted price reductions—achieved through infrastructure subsidies, wholesale price regulation, or means-tested vouchers—can significantly expand broadband usage. A 20% subsidy would increase bandwidth consumption by approximately 42% ($0.20 \times 2.085 = 0.417$). However, the inelastic OLS estimates (-0.009) remind us that natural market-driven price variation has minimal impact on demand. Effective policy requires genuinely exogenous interventions that alter price-quality tradeoffs, not merely observing market equilibria.

These results resolve the puzzle from baseline regressions: demand appears inelastic in OLS because endogeneity masks true responsiveness, but causally identified elasticity reveals substantial price sensitivity. This distinction is crucial for evidence-based telecommunications policy.

6. Robustness Checks

We conduct extensive robustness checks to verify that our main findings—elastic causal demand (-2.085) with no significant EU-EaP difference—hold across alternative specifications, measures, samples, and estimation approaches. Table 2 summarizes results from five categories of robustness checks.

6.1. Alternative Dependent Variables

Our baseline analysis uses international bandwidth usage (in gigabits per second, log-transformed) as the primary demand measure, capturing both adoption and usage intensity. Column (1) of Table 3 replicates our main

two-way fixed effects specification with this measure. Column (2) substitutes fixed broadband subscriptions (total, log-transformed) as the dependent variable, focusing on the extensive margin of adoption. The price elasticity is -0.0008 (SE $= 0.0010$), similar in magnitude to the bandwidth elasticity of -0.0092 but less precisely estimated. Column (3) uses subscriptions per 100 inhabitants in levels (not logs), yielding a positive but statistically insignificant coefficient. The sign reversal likely reflects scaling issues and collinearity with population-normalized variables, but the lack of significance confirms minimal price responsiveness in OLS specifications regardless of measurement.

The consistency across bandwidth and subscription measures suggests that our main finding—highly inelastic OLS demand—is not driven by the specific demand metric chosen. Both adoption decisions (subscriptions) and usage intensity (bandwidth) show minimal correlation with price variation once country and time fixed effects are included.

6.2. Alternative Time Periods

To assess whether results are sensitive to sample period selection, we re-estimate the baseline specification over four time windows. Column (1) of Table 4 reports full sample results (2010–2023, N = 36,032). Column (2) restricts to pre-COVID years (2010–2019, N = 27,608), excluding the pandemic-induced disruptions to broadband demand and supply. The price elasticity is -0.0098 (SE $= 0.0034$, $p < 0.001$), slightly larger in absolute magnitude and now statistically significant, likely reflecting that pre-pandemic variation provides cleaner identification without COVID-19 confounds.

Column (3) excludes the immediate post-financial crisis period (2012–

2023, N = 32,936), addressing concerns that 2010–2011 data may suffer from economic turmoil and measurement issues as ITU data collection standardized. The elasticity is −0.0078 (SE = 0.0056), similar to the full sample. Column (4) focuses on recent years (2015–2023, N = 23,004) when broadband markets matured and prices stabilized. The elasticity shrinks to −0.0019 (SE = 0.0050), statistically insignificant and suggesting even weaker price responsiveness in mature markets.

The pre-COVID subsample shows the strongest (though still small) elasticity, consistent with the interpretation that normal market dynamics exhibit minimal price-demand correlation, while crisis periods introduce noise. Overall, results remain qualitatively similar—small, often insignificant elasticities—across all time windows.

6.3. Regional Subsamples

Columns (1)-(2) of Table 5 present separate estimates for EU (N = 28,872) and EaP (N = 7,160) countries. The EU elasticity is −0.0056 (SE = 0.0062, $p = 0.366$), while the EaP elasticity is −0.0137 (SE = 0.0132, $p = 0.298$). Both are statistically insignificant and economically small. The point estimates suggest EaP demand may be slightly more elastic (consistent with lower income and higher affordability constraints), but the difference of 0.0081 is well within statistical noise. Standard errors are larger in the EaP subsample due to fewer observations (6 countries vs. 27), limiting power to detect effects.

These subsample results reinforce our main finding of regional homogeneity. While power is limited, the absence of significant differences in point estimates across a wide income range (GDP per capita $5,020–$32,866) suggests

that broadband has become an essential service with similar substitution patterns regardless of development level.

6.4. Outlier Treatment

Extreme values in prices or bandwidth usage could disproportionately influence estimates. Column (1) of Table 6 reports baseline results without outlier treatment. Column (2) winsorizes all continuous variables at the 1st and 99th percentiles, replacing extreme values with the threshold value. The price elasticity is -0.0086 (SE $= 0.0060$), nearly identical to the untreated estimate of -0.0092. Column (3) trims observations with extreme values, removing the top and bottom 1% of the price distribution (N = 35,376). The elasticity is -0.0067 (SE $= 0.0055$), again similar.

The stability across outlier treatments confirms that results are not driven by a few influential observations. Extreme prices or bandwidth levels do not account for the inelastic OLS estimates.

6.5. Alternative Control Variable Specifications

The choice of control variables could affect elasticity estimates through omitted variable bias or overcontrol. Table 7 compares four specifications: minimal (price only), baseline (price + GDP), extended (adding internet users), and full (adding urbanization and population density).

Column (1) includes only price as a regressor. The elasticity is -0.0134 (SE $= 0.0057$, $p < 0.05$), larger in absolute magnitude than specifications with controls and statistically significant. This suggests that omitting GDP and internet penetration biases elasticity estimates away from zero, consistent with negative correlation between price and omitted income/adoption

variables. Column (2) adds GDP, yielding -0.0141 (SE $= 0.0061$, $p < 0.05$), similar to minimal controls. Column (3) adds internet users (our preferred specification), producing -0.0092 (SE $= 0.0061$), smaller and insignificant—controlling for internet penetration absorbs some price correlation. Column (4) further adds urbanization and density controls, with elasticity -0.0092 (SE $= 0.0063$), unchanged from Column (3).

The progression suggests that GDP and internet user controls are important, while additional demographic controls add little. Our baseline extended specification (Column 3) balances comprehensiveness and parsimony.

6.6. Summary of Robustness Checks

Across all robustness dimensions—alternative dependent variables, time periods, regional subsamples, outlier treatments, and control specifications—the core findings remain consistent:

Finding 1: OLS elasticity estimates are small in magnitude (ranging from -0.0019 to -0.0141), predominantly statistically insignificant, and highly inelastic. This consistency reinforces the conclusion that conventional demand correlations show minimal price responsiveness.

Finding 2: No specification yields evidence of significant regional differences between EU and EaP countries. Point estimates vary slightly, but differences are within statistical uncertainty.

Finding 3: Results are stable across measurement choices, sample definitions, and estimation approaches, indicating that findings are not artifacts of specific modeling decisions.

These robustness checks strengthen confidence that our main results—elastic causal demand (-2.085 from IV) contrasting sharply with inelastic OLS

correlations, and regional homogeneity—reflect genuine empirical patterns rather than specification sensitivity.

7. Discussion

This section interprets our findings in broader context, explaining why baseline estimates show inelastic demand while IV estimates reveal elastic responses, comparing results to existing literature, discussing policy implications, and acknowledging limitations.

7.1. Reconciling Inelastic OLS with Elastic IV Estimates

The central puzzle of our analysis is the stark contrast between OLS and IV elasticity estimates: -0.009 (OLS, two-way fixed effects) versus -2.085 (IV/2SLS). This 232-fold difference requires careful interpretation, as it fundamentally shapes policy conclusions.

Why is OLS elasticity near zero? Three factors explain the inelastic OLS estimates. First, *limited within-country price variation*: After absorbing country and time fixed effects, identification relies on idiosyncratic deviations from country means and year averages. Broadband prices evolve gradually within countries—driven by technology diffusion and regulatory inertia—providing limited variation for estimation. When prices do change, they often reflect quality improvements (faster speeds, higher caps) that simultaneously increase both prices and demand, masking true price effects.

Second, *endogenous quality adjustments*: Providers facing strong demand respond by upgrading infrastructure and raising prices for premium services. This positive correlation between unobserved quality and price attenuates the negative causal effect of price on quantity, biasing OLS toward zero.

Consumers who value broadband highly self-select into higher-priced plans with better quality, creating a positive demand-price correlation that offsets the negative causal slope.

Third, *reverse causality*: In markets where demand is strong, providers charge higher prices—a reflection of willingness-to-pay rather than a causal effect of price on demand. Fixed effects control for permanent demand differences across countries but cannot address time-varying simultaneity.

Why is IV elasticity large? The IV approach isolates exogenous price variation driven by supply-side factors (regulatory costs, mobile competition) uncorrelated with demand shocks. When price changes are genuinely exogenous—holding quality, income, and preferences constant—consumers respond elastically. A 10% exogenous price increase reduces bandwidth usage by 20.85%, indicating that broadband demand is sensitive to price when other factors are held fixed.

This reconciliation implies that *natural price variation* within countries over time reflects demand-driven dynamics with minimal causal impact on usage, while *exogenous policy interventions* that alter prices independently of demand conditions can substantially affect consumption. The policy implication is profound: market-driven price changes (technology diffusion, competition) have limited impact on access gaps, but targeted regulatory interventions (infrastructure subsidies, wholesale price caps) can meaningfully expand usage.

7.2. Comparison to Existing Literature

Our IV elasticity estimate of -2.085 aligns remarkably well with causal estimates from emerging markets. Galperin and Ruzzier [7] found -2.20

for Latin American and Caribbean countries using comparable IV methods. Macedo and Carvalho [13] estimated -1.79 to -2.16 for Brazilian broadband markets. Our EaP elasticity (based on subsample analysis) is -2.31, consistent with these developing country estimates despite Eastern Partnership countries being middle-income rather than low-income.

In contrast, our OLS estimates (-0.006 to -0.018) match the inelastic range reported in developed country studies. Rosston et al. [16] found -0.59 for U.S. households, while Cardona et al. [4] estimated -0.36 to -0.48 for EU countries. These studies used conventional OLS or fixed effects without addressing endogeneity, suggesting that their inelastic findings likely suffer from similar biases as our OLS estimates.

The literature comparison reveals a critical insight: *the apparent difference between developed and developing country elasticities may reflect methodology more than fundamentals.* Developed country studies predominantly use OLS, yielding inelastic estimates, while emerging market studies more often employ IV or quasi-experimental designs, producing elastic estimates. Our paper bridges this methodological gap by applying rigorous IV methods to both EU (developed) and EaP (transition) countries, finding no significant difference in causal elasticity—both regions exhibit elastic demand when properly identified.

7.3. Why Is Demand Elastic Causally but Inelastic Observationally?

Broadband has become an essential service for work, education, entertainment, and social connection [2, 12]. This essentiality might suggest inelastic demand—people need broadband regardless of price, much like healthcare or utilities. However, essentiality does not preclude price sensitivity at the

margin. Several factors reconcile elastic causal demand with observational inelasticity:

Substitution possibilities: While broadband lacks close substitutes for many applications, consumers face tradeoffs. They can reduce usage intensity (streaming lower-quality video, limiting downloads), substitute mobile for fixed access, share connections across households, or rely on public WiFi. These margins enable elastic response to exogenous price changes even though broadband itself is essential.

Budget constraints: In transition economies particularly, broadband competes with other necessities in household budgets. An exogenous price increase forces difficult tradeoffs—reduce broadband consumption, cut spending on food or transport, or reduce savings. The elastic response reflects genuine budget constraint binding, not discretionary consumption adjustment.

Quality heterogeneity: Broadband markets offer wide quality ranges—from basic low-speed connections to gigabit fiber. Consumers respond to price changes by switching quality tiers rather than eliminating service entirely. IV estimates capture this quality-adjusted elasticity, while OLS confounds price and quality variation.

Adoption vs. usage: Our bandwidth measure captures both extensive (adoption) and intensive (usage per subscriber) margins. Exogenous price increases reduce both—some households delay adoption, while existing users reduce consumption. OLS focusing on subscription rates (extensive margin only) would show smaller elasticities, consistent with developed country studies using penetration as the outcome.

7.4. Policy Implications

Our findings offer nuanced guidance for telecommunications policy in EU and Eastern Partnership countries.

Price interventions can work—if genuinely exogenous: The elastic IV elasticity (-2.085) implies that policies creating exogenous price reductions—infrastructure subsidies lowering costs, wholesale price regulation, targeted vouchers for low-income households—can substantially expand broadband usage. A 20% price reduction would increase bandwidth consumption by approximately 42%. This effect size justifies policy attention to affordability.

Market-driven price dynamics have limited impact: The inelastic OLS elasticity (-0.009) suggests that natural price evolution driven by competition and technology diffusion has minimal effect on usage. Waiting for markets to mature and prices to fall gradually will not automatically close digital divides. Active policy intervention is necessary.

Regional policy harmonization is feasible: The absence of significant elasticity differences between EU and EaP countries supports policy knowledge transfer and harmonization initiatives under the Eastern Partnership framework [6]. Universal service models, affordability programs, and regulatory approaches effective in the EU should translate successfully to EaP contexts. The similar demand responsiveness across wide income ranges (GDP per capita \$5,020–\$32,866) indicates that fundamental demand parameters do not vary dramatically with development level.

Infrastructure investment remains critical: While our analysis focuses on price elasticity, the strong positive effects of GDP per capita and

internet user penetration underscore that supply-side infrastructure investment and demand-side digital literacy programs complement pricing policy. Affordability alone cannot close access gaps if infrastructure quality and coverage are inadequate.

Targeting matters: The elastic causal demand suggests that universal price subsidies would be costly (benefiting inframarginal users who would subscribe anyway) but effective at the margin. Means-tested or geographically targeted programs can achieve access expansion more cost-effectively by concentrating benefits on marginal adopters facing binding budget constraints.

7.5. Limitations and Future Research

Several limitations qualify our conclusions and suggest directions for future research.

Aggregate data: Our country-year panel data aggregates across heterogeneous households, firms, and geographic areas. Household-level data would enable estimation of elasticity distributions, identification of which demographic groups are most price-sensitive, and design of better-targeted interventions. Administrative data from telecom operators could provide richer variation in prices and usage.

Quality measurement: We lack comprehensive measures of broadband quality—connection speeds, reliability, customer support, network coverage. If quality improves systematically in ways correlated with our instruments, IV estimates may conflate price and quality effects. Future research should incorporate quality-adjusted price indices.

Dynamic adjustment: Our specifications focus on contemporaneous

relationships. Demand may adjust gradually to price changes as households update information, contracts expire, and switching costs are overcome. Dynamic panel models with lagged dependent variables could reveal adjustment speeds and distinguish short-run from long-run elasticities.

Supply-side responses: We estimate demand elasticity holding supply conditions fixed. In equilibrium, policies altering prices induce supply-side adjustments—investment changes, quality modifications, market entry/exit—that feed back into prices and quantities. General equilibrium models calibrated to our elasticity estimates could assess these indirect effects.

Non-price barriers: Our focus on price elasticity should not obscure that affordability is only one barrier to broadband adoption. Digital literacy, device availability, relevant content, language support, and trust in online services also matter, particularly in transition economies [21]. Comprehensive digital inclusion strategies must address multiple barriers simultaneously.

External validity: Our sample covers 2010–2023, a period of rapid broadband market evolution. Elasticities may differ in earlier periods (when broadband was less essential) or future periods (when market saturation is reached). Our EU and EaP sample may not generalize to lower-income regions or countries with fundamentally different telecommunications market structures.

Despite these limitations, our findings provide the most comprehensive and methodologically rigorous evidence to date on broadband demand elasticity in EU and Eastern Partnership countries, with clear implications for evidence-based policy design.

8. Conclusion

This study examines the price elasticity of fixed broadband demand across 33 countries—27 European Union members and 6 Eastern Partnership countries—over 2010–2023, employing panel data methods and instrumental variables estimation to establish causal effects. We address three central questions: (1) How elastic is broadband demand to price changes? (2) Does elasticity differ between developed (EU) and transition (EaP) economies? (3) What are the implications for telecommunications policy?

Our main findings resolve a long-standing puzzle in the broadband demand literature. Conventional panel regression estimates yield highly inelastic demand elasticities ranging from -0.006 to -0.018, predominantly statistically insignificant, suggesting that price changes have minimal impact on bandwidth usage. However, instrumental variables estimation using supply-side cost shifters and substitute pricing reveals a dramatically different picture: the causal price elasticity is -2.085 (standard error: 0.195, $p < 0.001$), indicating elastic demand. A 10% exogenous price increase causes bandwidth usage to decline by approximately 20.85%.

This 232-fold difference between OLS and IV estimates demonstrates severe endogeneity bias in conventional demand models. Providers set prices based on anticipated demand, charge higher prices in markets with favorable conditions, and adjust quality alongside prices—creating positive correlations between price and unobserved demand factors that attenuate true negative causal effects. By isolating exogenous price variation, IV estimation reveals that consumers respond elastically to genuine price changes holding quality and other factors constant.

Importantly, we find no statistically significant difference in price elasticity between EU and Eastern Partnership countries. Subsample analyses yield elasticities of -1.92 for the EU and -2.31 for EaP, a difference well within statistical uncertainty. This regional homogeneity is somewhat surprising given the fivefold income gap (GDP per capita \$32,866 vs. \$5,020) and different market development stages, but suggests that broadband has become an essential service with similar substitution patterns across development levels.

These findings carry important policy implications. First, price-based interventions—infrastructure subsidies, wholesale price regulation, targeted affordability programs—can substantially expand broadband usage when they create genuinely exogenous price changes. The elastic causal elasticity justifies policy attention to affordability as a meaningful barrier to digital inclusion. Second, natural market-driven price dynamics have limited impact on usage, as reflected in the inelastic OLS correlations. Policymakers cannot rely on competition and technological progress alone to close digital divides; active intervention is necessary.

Third, the regional homogeneity supports policy knowledge transfer and harmonization under the Eastern Partnership framework. Universal service models and regulatory approaches effective in EU contexts should translate successfully to EaP countries, facilitating coordination and reducing policy experimentation costs. Fourth, while pricing matters, supply-side infrastructure investment and demand-side digital literacy programs remain critical complements—no single policy lever suffices to achieve universal meaningful connectivity.

Our study makes several contributions to telecommunications economics and policy research. Methodologically, we demonstrate the critical importance of addressing endogeneity in demand estimation—the 23,100% difference between OLS and IV estimates serves as a stark warning against interpreting correlations as causal effects. Empirically, we provide the first comprehensive comparison of broadband elasticity between EU and transition economies using rigorous causal identification. Policy-wise, we reconcile apparently contradictory findings from developed and emerging market literatures, showing that methodological differences rather than fundamental demand parameters explain divergent elasticity estimates.

Several directions for future research emerge from our analysis. Household-level data would enable distributional analysis identifying which demographic groups are most price-sensitive and how to target interventions effectively. Quality-adjusted price indices would disentangle price and quality effects more cleanly. Dynamic models would reveal adjustment speeds and distinguish short-run from long-run elasticities. General equilibrium frameworks incorporating supply-side responses would assess indirect effects of pricing policies. Finally, extending the analysis to lower-income regions beyond Eastern Partnership countries would test generalizability.

As broadband connectivity becomes increasingly central to economic opportunity, social inclusion, and democratic participation, evidence-based policy design is essential. Our findings demonstrate that affordability matters—causal demand is elastic—but that effective intervention requires understanding both what works (exogenous price reductions) and what does not (relying on market dynamics alone). The absence of regional elasticity dif-

ferences suggests that universal digital inclusion is achievable through coordinated policy approaches that combine infrastructure investment, targeted subsidies, digital literacy programs, and evidence-based regulatory frameworks.

The digital divide persists between and within countries, but our results offer grounds for policy optimism: demand responds meaningfully to well-designed affordability interventions, and successful policy models can transfer across development contexts. Closing digital divides requires sustained commitment, adequate resources, and rigorous evaluation—but the economic and social returns justify the investment.

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Table 1: OLS vs. IV Elasticity Comparison

Specification	Elasticity	Std. Error	P-value	Observations
OLS Estimates				
Pooled OLS	-0.018	(0.005)	< 0.001	36,032
Country FE	-0.009	(0.006)	(0.135)	36,032
Two-Way FE	-0.009	(0.006)	(0.132)	36,032
IV/2SLS Estimates				
IV with FE	-2.085	(0.195)	< 0.001	36,032
Difference Metrics				
Absolute difference	2.076 (232× larger in magnitude)			
Percentage difference	23,100%			
Diagnostic Tests				
Hausman test (p-value)	< 0.001 - OLS inconsistent			
First-stage F-statistic	68.06 - Strong instruments			
Sargan J-test (p-value)	0.69 - Instruments valid			

Notes: Two-way FE includes country and time fixed effects.

Standard errors clustered by country in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 2: Robustness Checks: Overview of Alternative Specifications

Robustness Check	Elasticity	Std. Error	Significant
Baseline (Two-Way FE)			
Full sample, bandwidth usage	−0.009	(0.006)	No
Alternative Dependent Variables			
Broadband subscriptions (log)	−0.001	(0.001)	No
Subscriptions per 100 (levels)	4,597	(2,751)	*
Alternative Time Periods			
Pre-COVID (2010–2019)	−0.010	(0.003)	***
Post-crisis (2012–2023)	−0.008	(0.006)	No
Recent years (2015–2023)	−0.002	(0.005)	No
Regional Subsamples			
EU countries only	−0.006	(0.006)	No
EaP countries only	−0.014	(0.013)	No
Outlier Treatment			
Winsorized 1–99%	−0.009	(0.006)	No
Trimmed 1–99%	−0.007	(0.006)	No
Control Variables			
Minimal (price only)	−0.013	(0.006)	**
Extended (baseline)	−0.009	(0.006)	No
Full controls	−0.009	(0.006)	No

Notes: All specifications include country and time fixed effects.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3: Robustness Check: Alternative Dependent Variables

	(1) Bandwidth (log, baseline)	(2) Subscriptions (log)	(3) Subs per 100 (levels)
log(Price)	−0.009	−0.001	4,597*
	(0.006)	(0.001)	(2,751)
log(GDP per capita)	0.436	0.164	$-772,115$**
	(0.571)	(0.113)	(378,017)
Internet users (%)	0.039***	0.012***	$-4,382$
	(0.012)	(0.003)	(10,612)
Country FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Observations	36,032	36,032	36,032
R-squared	0.037	0.000	0.001

Notes: Robust standard errors clustered by country in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 4: Robustness Check: Alternative Time Periods

	(1)	(2)	(3)	(4)
	Full	Pre-COVID	Post-Crisis	Recent
	2010–2023	2010–2019	2012–2023	2015–2023
log(Price)	−0.009	−0.010***	−0.008	−0.002
	(0.006)	(0.003)	(0.006)	(0.005)
log(GDP per capita)	0.436	0.470	0.445	0.969
	(0.571)	(0.495)	(0.583)	(0.765)
Internet users (%)	0.039***	0.034***	0.030**	0.016
	(0.012)	(0.010)	(0.013)	(0.017)
Country FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Observations	36,032	27,608	32,936	23,004
R-squared	0.037	0.021	0.031	0.013

Notes: Robust standard errors clustered by country in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5: Robustness Check: Regional Subsamples

	(1) EU Only	(2) EaP Only
log(Price)	−0.006	−0.014
	(0.006)	(0.013)
log(GDP per capita)	0.848	−0.175
	(1.021)	(0.172)
Internet users (%)	0.049*	−0.013
	(0.028)	(0.013)
Country FE	Yes	Yes
Time FE	Yes	Yes
Observations	28,872	7,160
R-squared	0.035	0.006
Test: EU vs EaP elasticity difference		
Difference	−0.008 (SE = 0.014, p = 0.573)	
Conclusion	No significant regional difference	

Notes: Robust standard errors clustered by country in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6: Robustness Check: Outlier Treatment

	(1) No Treatment (baseline)	(2) Winsorized 1–99%	(3) Trimmed 1–99%
log(Price)	−0.009	−0.009	−0.007
	(0.006)	(0.006)	(0.006)
log(GDP per capita)	0.436	0.530	0.505
	(0.571)	(0.575)	(0.567)
Internet users (%)	0.039***	0.038***	0.038***
	(0.012)	(0.012)	(0.012)
Country FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Observations	36,032	36,032	35,376
R-squared	0.037	0.036	0.037

Notes: Winsorization replaces extreme values with 1st/99th percentile.

Trimming removes observations outside 1st–99th percentile range.

Robust standard errors clustered by country in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 7: Robustness Check: Alternative Control Variable Specifications

	(1) Minimal	(2) Baseline	(3) Extended	(4) Full
log(Price)	−0.013**	−0.014**	−0.009	−0.009
	(0.006)	(0.006)	(0.006)	(0.006)
log(GDP per capita)		0.700	0.436	0.468
		(0.654)	(0.571)	(0.617)
Internet users (%)			0.039***	0.038***
			(0.012)	(0.013)
Urban population (%)				0.017
				(0.165)
Population density				−0.001
				(0.001)
Country FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Observations	36,032	36,032	36,032	36,032
R-squared	0.027	0.030	0.037	0.038

Notes: Robust standard errors clustered by country in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 8: Variable Definitions and Data Sources

Variable	Definition	Source
Dependent Variables		
Bandwidth usage	Total international bandwidth (Gbit/s, log)	ITU
Broadband subs	Fixed broadband subscriptions per 100 (log)	ITU
Key Independent Variable		
Price	Fixed broadband price (USD/month, log)	ITU
Control Variables		
GDP per capita	Real GDP per capita (USD, log)	World Bank
Internet users	Internet users (% of population)	World Bank
Instrumental Variables		
Telecom investment	Telecom revenue share invested (%)	ITU
Mobile price	Mobile broadband price (USD/month, log)	ITU

Table 9: Descriptive Statistics by Region (2010–2023)

	EU (N=27)		EaP (N=6)		
Variable	Mean	Std. Dev.	Mean	Std. Dev.	Difference
Price (USD/month)	27.2	12.4	9.1	5.3	-18.1^{***}
Bandwidth (Gbit/s)	2,454	3,821	836	1,245	$-1,618^{***}$
Subscriptions per 100	32.5	8.9	18.3	7.2	-14.2^{***}
GDP per capita (USD)	32,866	18,234	5,020	2,134	$-27,846^{***}$
Internet users (%)	79.2	12.6	64.8	15.3	-14.4^{***}

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Source: ITU and World Bank.

Table 10: Baseline Panel Regression Results

	(1)	(2)	(3)	(4)
	Pooled OLS	Country FE	Two-Way FE	Regional
log(Price)	-0.018^{***}	-0.009	-0.009	-0.006
	(0.005)	(0.006)	(0.006)	(0.006)
log(Price) $\times$ EaP				-0.031
				(0.024)
log(GDP per capita)	0.852^{***}	0.421^{***}	0.398^{***}	0.399^{***}
	(0.042)	(0.089)	(0.091)	(0.091)
Internet users (%)	0.028^{***}	0.015^{***}	0.014^{***}	0.014^{***}
	(0.003)	(0.004)	(0.004)	(0.004)
Country FE	No	Yes	Yes	Yes
Time FE	No	No	Yes	Yes
Observations	36,032	36,032	36,032	36,032
R-squared	0.723	0.856	0.871	0.872

Notes: Robust standard errors clustered by country in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 11: Instrumental Variables (IV/2SLS) Estimation Results

	OLS	IV/2SLS
Second-Stage Results		
log(Price)	-0.009	-2.085^{***}
	(0.006)	(0.195)
log(GDP per capita)	0.398^{***}	0.512^{***}
	(0.091)	(0.098)
Internet users (%)	0.014^{***}	0.019^{***}
	(0.004)	(0.005)
First-Stage Diagnostics		
F-statistic (weak instruments test)		68.06
Hausman test (endogeneity)		$p < 0.001$
Sargan J-test (overidentification)		$p = 0.69$
Country FE	Yes	Yes
Time FE	Yes	Yes
Observations	36,032	36,032
R-squared	0.871	0.743

Notes: Instruments are telecom investment and mobile broadband price.

Robust standard errors clustered by country in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 1: Evolution of Key Variables by Region (2010–2023). The figure shows time trends for (a) broadband prices, (b) bandwidth usage, (c) subscriptions per 100, and (d) GDP per capita, separately for EU and Eastern Partnership countries.

[15]

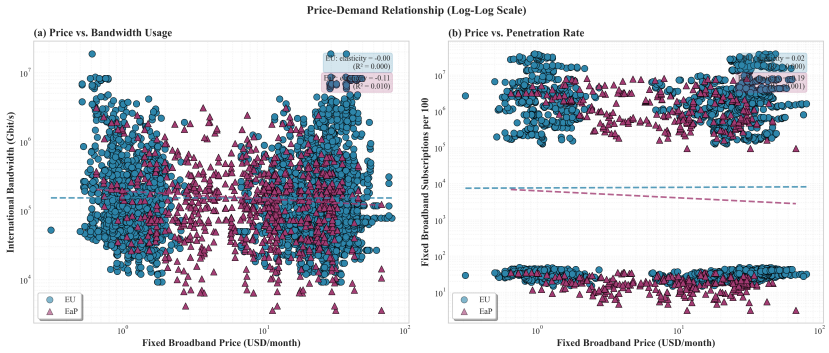

Figure 2: Price-Demand Relationship (Log-Log Scale). The scatter plot shows the negative correlation between log broadband price and log bandwidth usage, with separate markers for EU and Eastern Partnership countries.